ch10 Projectile Motion

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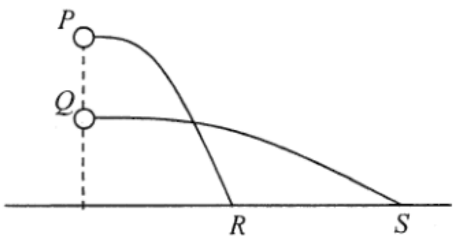
2017 Q9

*9. Marbles P and Q of the same mass are shot horizontally. They hit the horizontal ground at points R and S respectively as shown. Neglect air resistance.

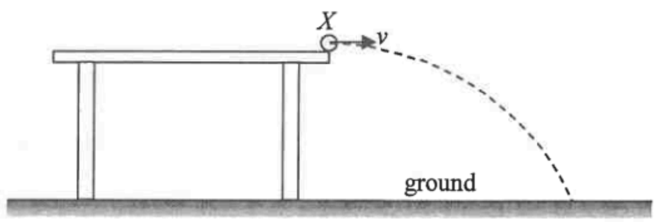


Which of the following statements is **INCORRECT** ?

A. The initial speed of marble P is smaller than that of marble Q.
B. The time of flight of marble P is shorter than that of marble Q.
C. The potential energy loss of marble P is greater than that of marble Q.
D. The acceleration of marbles P and Q is the same during the flight.

2022 Q13

*13. A marble leaves the edge of a table at point X with a horizontal speed v and hits the ground at a certain point as shown.



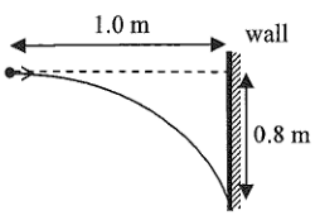
If the marble leaves the table at a higher speed, which of the following would remain unchanged ? Neglect air resistance.

(1) the time of flight of the marble in the air
(2) the acceleration of the marble during its flight in air
(3) the horizontal distance between X and the landing point

A. (1) only
B. (3) only
C. (1) and (2) only
D. (2) and (3) only

2014 Q10

*10.

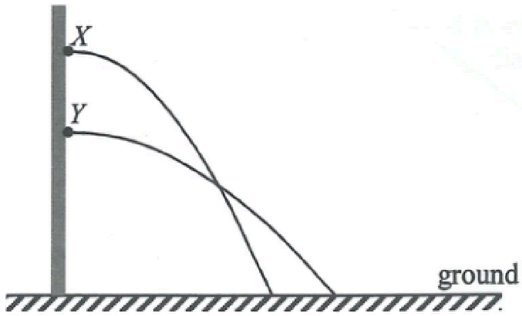


A particle is projected horizontally towards a vertical wall 1.0 m away. It hits the wall at a position 0.8 m vertically below its point of projection. At what speed is it projected ? Neglect air resistance. ($g = 9.81$ m s^{-2})

A. 2.0 m s^{-1}
B. 2.5 m s^{-1}
C. 5.0 m s^{-1}
D. 6.3 m s^{-1}

2019 Q13

*13.



Particles X and Y are projected horizontally from a vertical wall and the figure shows their paths in air before reaching the ground. Which statements below are correct ? Neglect air resistance.

(1) The time of flight of Y is longer.
(2) The projection speed of Y is greater.
(3) X and Y can have the same landing speed.

A. (1) and (2) only
B. (1) and (3) only
C. (2) and (3) only
D. (1), (2) and (3)

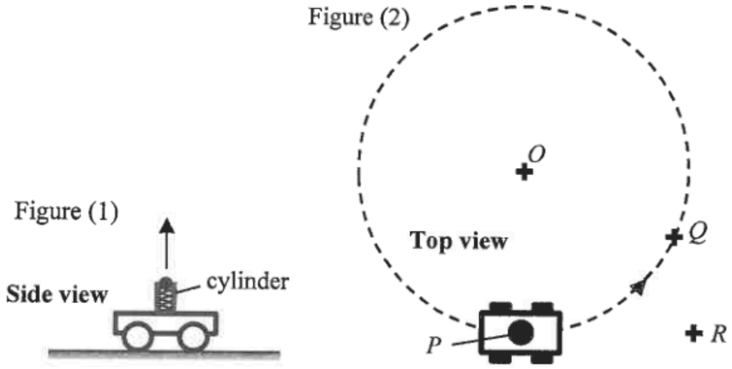
2012 Q12

*12. A bomber aircraft is 1 km above the ground and is flying horizontally at a speed of 200 m s^{-1}. The aircraft is going to release a bomb to destroy a target on the ground. How long before flying over the target should the bomb be released ? Assume that the bomber aircraft and the target are in the same vertical plane and neglect air resistance. ($g = 9.81$ m s^{-2})

A. 5.6 s
B. 10.1 s
C. 14.3 s
D. It cannot be calculated as the horizontal distance between the aircraft and the target is not known.

2022 Q9

*9. A trolley with a spring-loaded launcher in Figure (1) can project a ball vertically upward. The trolley travels on the ground at a constant speed in a horizontal circle (with centre O) as shown in Figure (2).

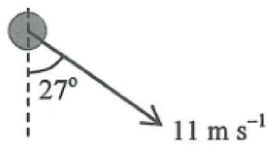


The ball is projected when the trolley is at point P. After some time, the ball falls back to the ground as the trolley just reaches point Q. Neglecting air resistance, where would the ball land ?

A. O
B. P
C. Q
D. R

2019 Q6

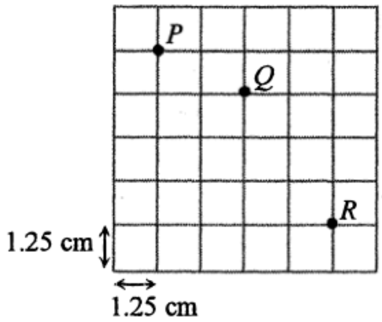
*6. A small ball after projection moves under the effect of gravity only. Its velocity at a certain instant is shown below. What is the speed of the ball **1 s before** ? Neglect air resistance. ($g = 9.81$ m s^{-2})



A. 19.1 m s^{-1}
B. 9.8 m s^{-1}
C. 5.0 m s^{-1}
D. 0.2 m s^{-1}

2016 Q10

*10.

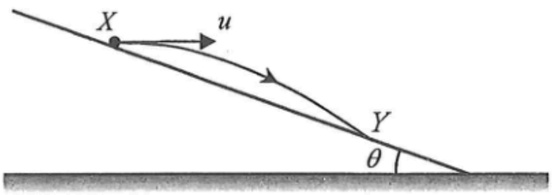


The above stroboscopic picture shows a particle projected horizontally at position P into the air in a vertical plane. Subsequently the particle reaches positions Q and R such that the time interval between P and Q is equal to that between Q and R. Each square of the grid measures 1.25 cm × 1.25 cm. Find the particle's speed of projection at P. Neglect air resistance. ($g = 9.81$ m s^{-2})

A. 0.3 m s^{-1}
B. 0.4 m s^{-1}
C. 0.5 m s^{-1}
D. 0.6 m s^{-1}

2025 Q9

*9.



X and Y are two points on an incline making an angle θ with the horizontal. A particle projected horizontally from X with speed u lands at Y as shown. Which expression correctly gives its time of flight from X to Y? Neglect air resistance.

A. $\dfrac{u \sin \theta}{2g}$

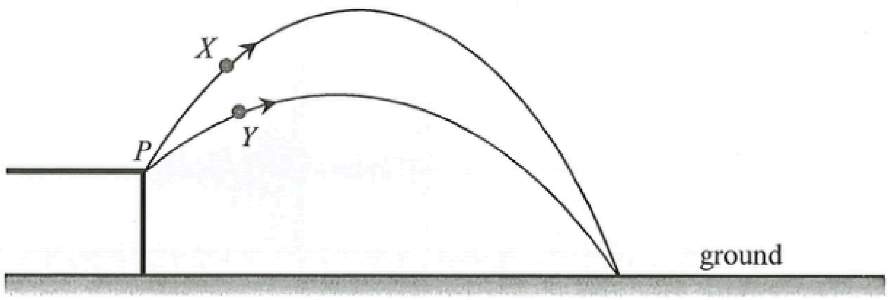
B. $\dfrac{u \tan \theta}{2g}$

C. $\dfrac{2u \sin \theta}{g}$

D. $\dfrac{2u \tan \theta}{g}$

2023 Q13

*13. Two identical particles X and Y are projected from point P **simultaneously with the same initial speed** but with different angles as shown. They finally hit the ground at the same point via different paths.



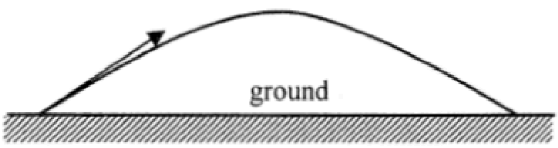
Which of the following statements is/are correct ? Neglect air resistance.

(1) They hit the ground at the same time.
(2) They hit the ground with the same speed.
(3) The kinetic energy of Y is greater than that of X at their respective maximum heights.

A. (1) only
B. (3) only
C. (1) and (2) only
D. (2) and (3) only

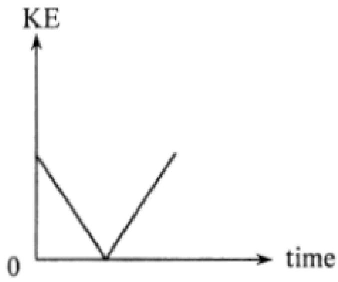
2013 Q13

*13.

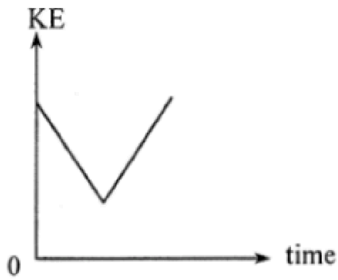


A particle is projected into the air at time $t = 0$ and it performs a parabolic motion before landing on the ground as shown. Which graph represents the variation of the kinetic energy (KE) of the particle with time before landing ? Neglect air resistance.

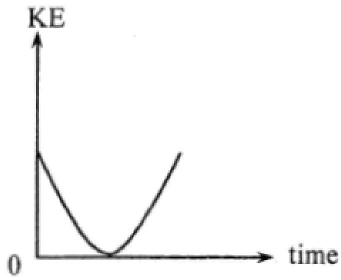
A.



B.



C.



D.